



ELE101

SDR&DSP&Decibel

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Outline

- Quick review
- SDR & Digital Communications
- Decibel Calculation
- Python 101



Review

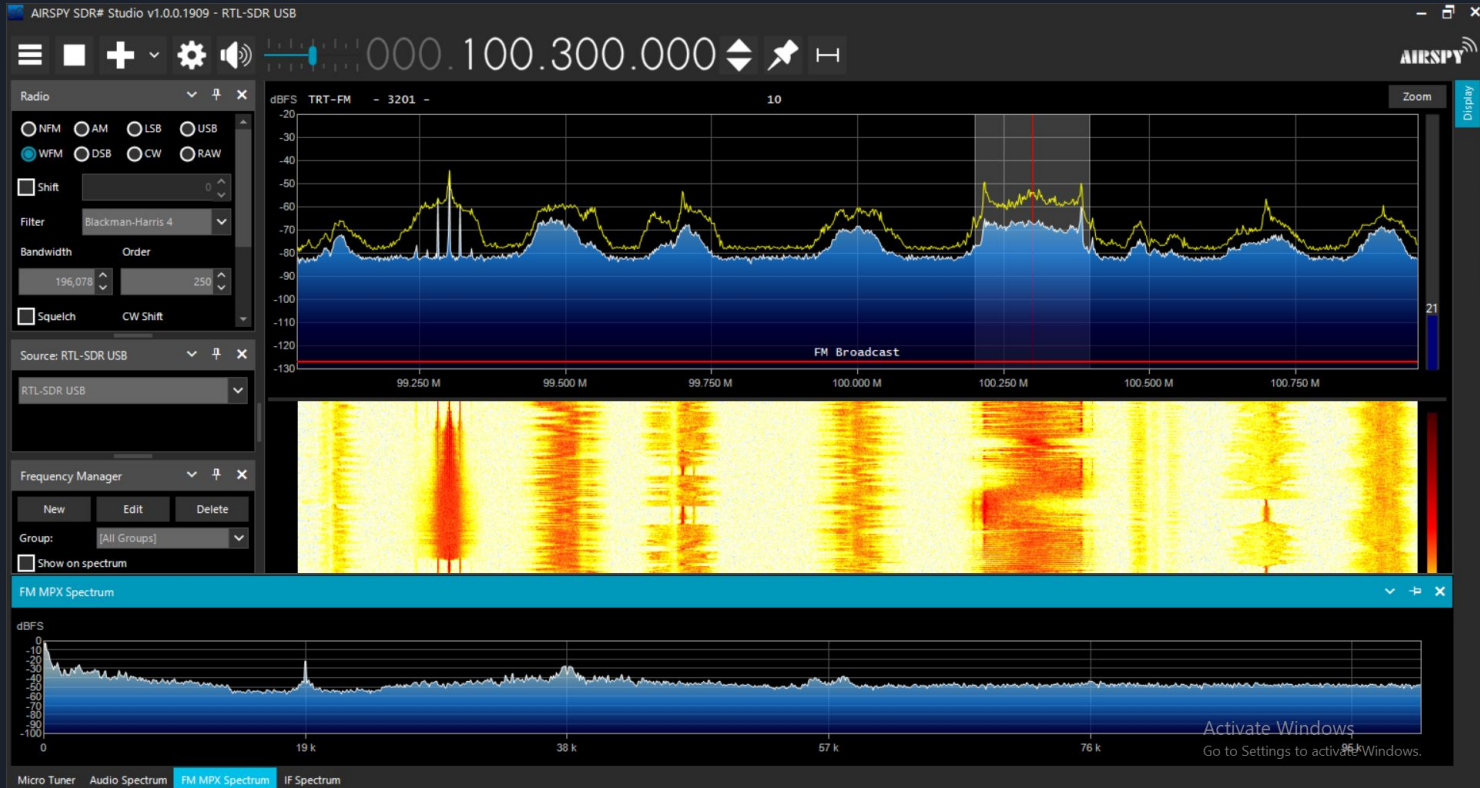
- Git-GitHub
- Linux
- miniconda
- Jupyter Lab
- Python
- Communication Systems



Assignment Submission

- Accept the assignment
- A new repo will be created for you, automatically
- Clone assignment repo into your PC
- change to your local directory (using cd command)
- Follow instructions in the README file
- Submit your assignment using git commands
 - `git add .`
 - `git commit - m "something message here"`
 - `git push`

Merhaba Dünya: WBFM





Communication Systems

- Described as Band, Frequency, or Wavelength
- Bands: VHF
- Frequency: 100 MHz
- Wavelength: 3 m
- $\lambda * \text{frequency} = \text{speed of light}$
 - $\lambda * f = c$
- Wavelength (in m) = $300 / (\text{frequency in MHz})$

Yazılım Tabanlı Radyo





YTR Özellikleri

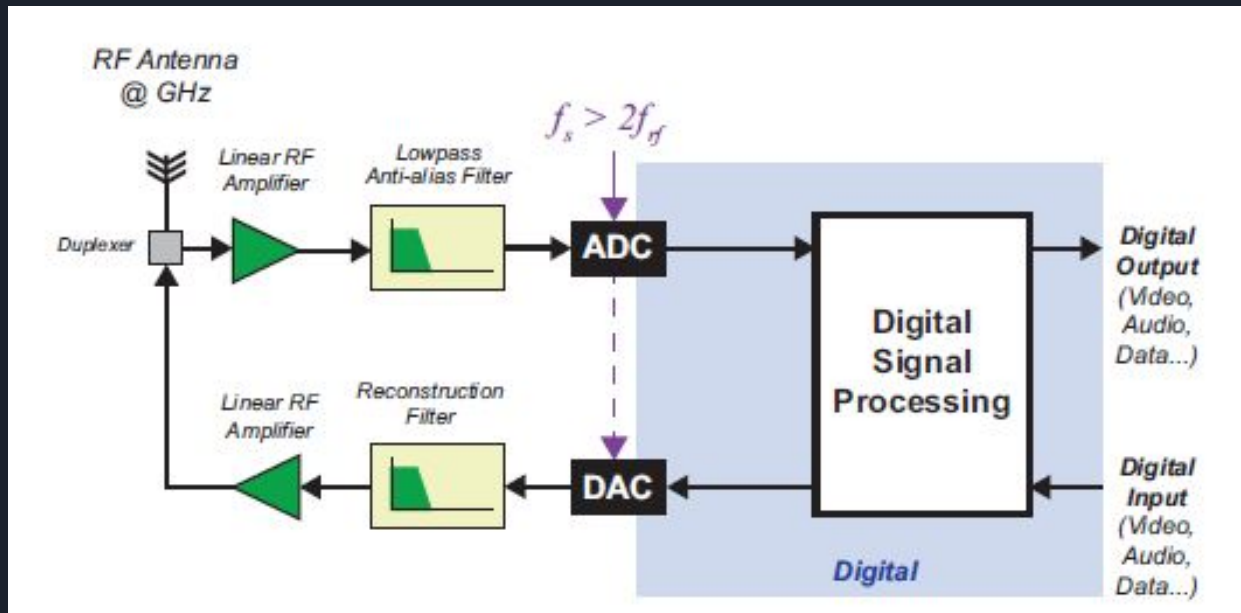
- Frekans Kapsaması
- Dar/Geniş anlık bant genişliği
Ara yüz (USB, Ethernet, PCIe)
- Rx/Tx, half-duplex, full-duplex, MIMO
- Preselektörler
- Bütçe: 50\$-...k\$



RTL-SDR

- 8-bit ADC
- 24MHz-1.7GHz (değişkenlik gösterebilir)
- 2.4Msps BW (kararlı) 3.2MHz'e kadar
- “*HamItUp*” upconverter
 - HF kapsam

YTR Mimarisi

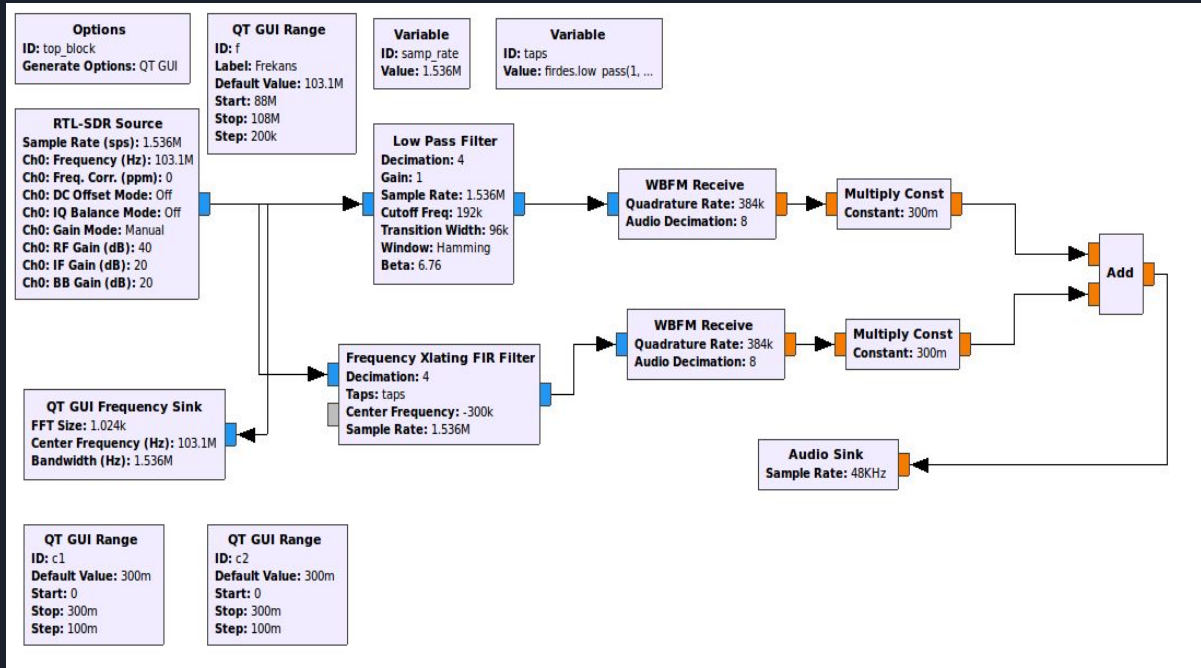




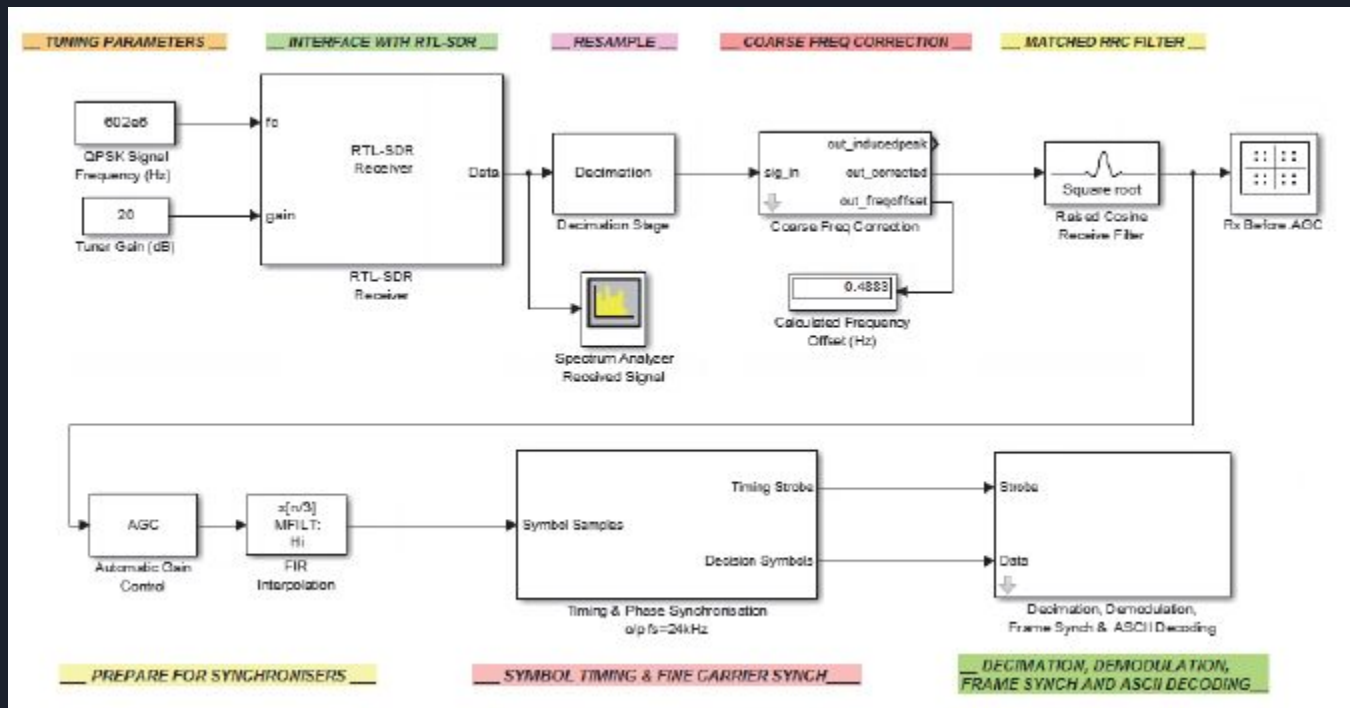
SDR Ortamları

- Hazır uygulamalar
 - SDR# (Windows)
 - gqrx (Linux, Mac)
 - SDR++, SDRAngel
- Kod tabanlı
 - C++, MATLAB, Python
- Blok tabanlı
 - GNU Radio, Simulink, LabView

GNU Radio



Simulink





Projeler, projeler ...

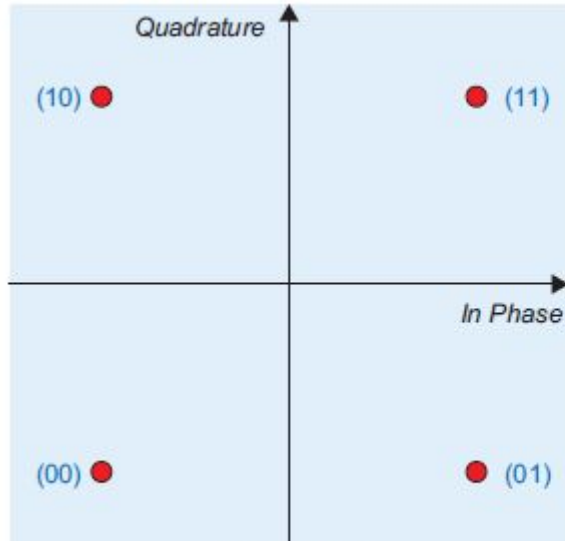
- Hava Trafik Kontrolü
- Uçaklar, ADS-B
- Gemiler, AIS
- Hava Durumu Balonları/Uyduları
- Lastik basıncı
- Tersine Mühendislik: Medikal cihazlar
- Pasif Radar
- Yön/Konum Bulma



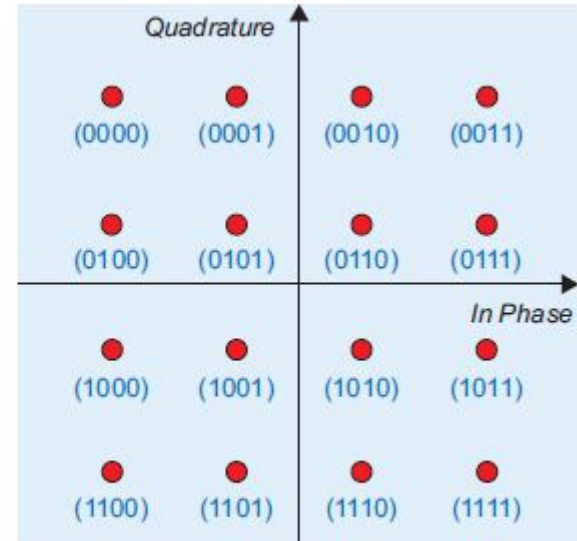
Sayısal Haberleşme

- Gürültüye Dayanıklılık
- Sıkıştırma ve Verimlilik
- Güvenlik
- Tekrar Oluşturma ve Kaydetme

Sayısal Modülasyon



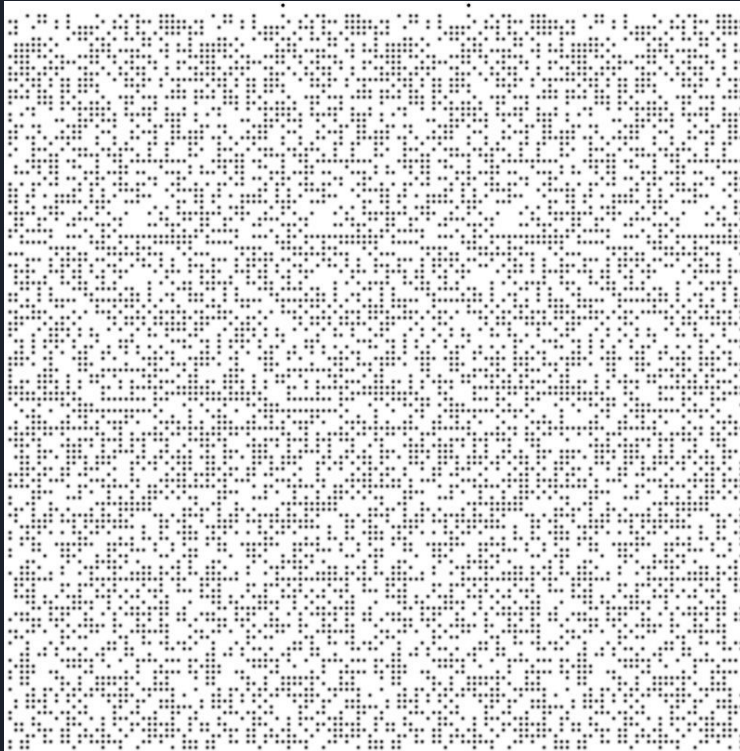
(a) 4-QAM



(b) 16-QAM



16384 QAM





Mobilite

- 1990'lar mesajlaşma, 2G, kbps
- 2000s 3G, 100kbps, WiFi
- Bugün: LTE, WiFi, MIMO
- Yarın: 5G, IoT
- Akıllı Telefonlar (velcro yaklaşımı, ~8 radios)
 - GSM (900M)
 - GPS (1.5G)
 - WiFi, Bluetooth (2.4G)
 - LTE (800M)
 - UMTS (2.1G)



WaveLAN, the starting point for Wi-Fi development, was used for wirelessly connecting cashing machines.

802.11b

Higher speed physical layer extension in the 2.4 GHz band

- Channel bandwidth: 22 MHz
- Modulation: DSSS
- Throughput: 11 Mbps
- Standard: IEEE 802.11b-1999

802.11g

High speed physical layer in the 2.4 GHz band

- Channel bandwidth: 22 MHz
- Modulation: OFDM
- Throughput: 54 Mbps
- Standard: IEEE 802.11g-2003

Need for faster speed and better distance coverage.

802.11n

Further higher data rate extension

- Channel bandwidth: 40 MHz
- Modulation: MIMO
- Throughput: 600 Mbps
- Standard: IEEE 802.11n-2009

The ability to connect to the internet via mobile devices and the rising number of smartphones on the market required the introduction of features like MIMO.

802.11n

Enhancement for higher throughput (HT)

- Channel bandwidth: 40 MHz
- Modulation: MIMO
- Throughput: 600 Mbps
- Standard: IEEE 802.11n-2009

More and more people wanted Wi-Fi at home and at work. High speed Wi-Fi was therefore required in the 5 GHz spectrum to relieve the overcrowded 2.4 GHz spectrum.

802.11ac

Enhancement for very high throughput (VHT)

- Channel bandwidth: 160 MHz
- Modulation: MIMO
- Throughput: 3.5 Gbps
- Standard: IEEE 802.11ac-2013

Designed for in-room desk network applications requiring very high data rates such as for HD video streaming.

The heavy use of Wi-Fi meant that a new approach was required. OFDMA allows multiple devices to communicate simultaneously.

802.11ax

Enhancement for high efficiency (HE) Wi-Fi

- Channel bandwidth: 160 MHz
- Modulation: MIMO
- Throughput: 9.6 Gbps
- Standard: IEEE 802.11ax-2021

The advent of home office and schooling as well as industrial applications require improved data throughput, reduced latency and efficiency.

Learn more about IEEE 802.11 testing: www.rohde-schwarz.com/wlan



Achieves up to 20 Gbps throughput and enables extended distances for enlarged application spaces.

802.11be

Enhancement (EHT) (EHT) in bands above 6 GHz

- Channel bandwidth: 320 MHz
- Modulation: MIMO
- Throughput: 460 Gbps
- Standard: IEEE 802.11be-2024

Multi-antenna transceiver methods

The evolution from SISO to single-user and multi-user MIMO was essential to meet data throughput demands.

Single input single output (SISO)

Use of a single antenna on access points and devices for sequential communication of the access point with connected devices, requiring a carrier sense multiple access (CSMA) scheme to control spectrum access.



Single-user multiple input multiple output (SU-MIMO)

Use of multiple antennas to improve data throughput, requiring a carrier sense multiple access (CSMA) scheme to control spectrum access.



Multi-user MIMO

Based on OFDMA, MU-MIMO allows simultaneous communications of stations in parallel. Scheduling enables multiple users to access individual MIMO antennas at the same time, increasing efficient spectrum utilization.



Test and measurement solutions from Rohde & Schwarz

R&S CMW710 wireless connectivity tester Threat radar expert designed for testing 5G E-UTRA access points (APs) and devices (UEs) in 5G and 4G-LTE networks.	R&S CMW710 communications manufacturing test set Ultra-compact non-signting test set for 5G E-UTRA access points (APs) and devices (UEs) in 5G and 4G-LTE networks.	R&S SMW710A vector signal generator Threat radar expert in its class with up to 500 MHz modulation bandwidth and real-time MIMO testing with real-time baseband.	R&S SMW710B vector signal generator Threat radar expert in its class with up to 500 MHz modulation bandwidth and real-time MIMO testing with real-time baseband.	R&S FSW signal and spectrum analyzer Setting standards in innovation and mobility by testing 5G E-UTRA access points (APs) and devices (UEs) in 5G and 4G-LTE networks.	R&S FSW300 signal and spectrum analyzer The right choice for 5G E-UTRA access points (APs) and devices (UEs) in 5G and 4G-LTE networks.	R&S TS2007 regulatory test system for wireless devices Testing of wireless devices according to the 3GPP standards in 5G and 4G-LTE networks.	R&S DT200 RF diagnostic chamber RF diagnostic chamber for RF analysis during development. Supports a wide range of calibrated test applications for 5G E-UTRA.
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Wi-Fi is a registered trademark of Wi-Fi Alliance

ROHDE & SCHWARZ
Make ideas real



The HISTORY and FUTURE of Wi-Fi



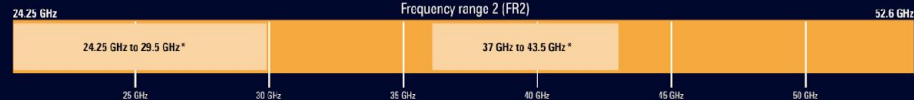


Hücresel Teknoloji

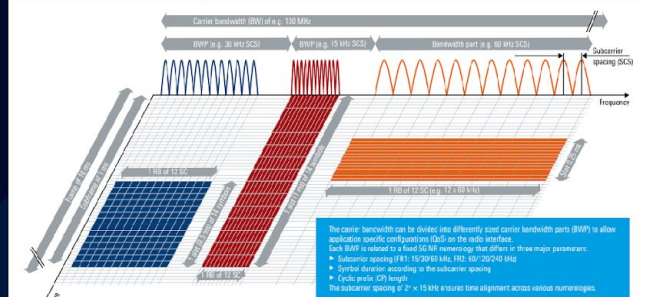
- 1G - 1979
 - Analog, voice only
- 2G - 1991
 - SMS, MMS, WAP
 - TDMA
 - 200k
- 3G UMTS - 1998
 - WCDMA
 - 5M
- 4G LTE - 2008
 - OFDMA
 - 1.4M-20M
- 5G - 2019

DEMYSTIFYING 5G NR

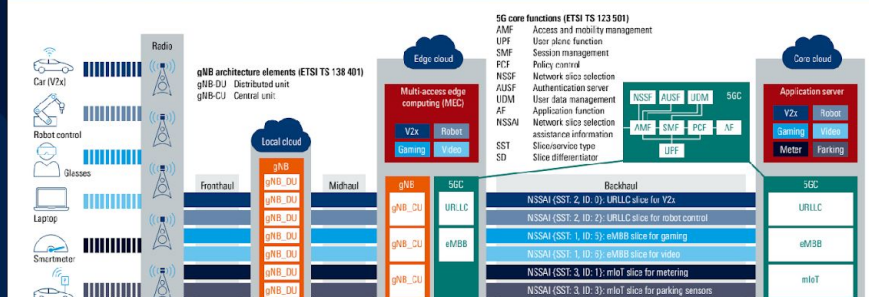
For more information



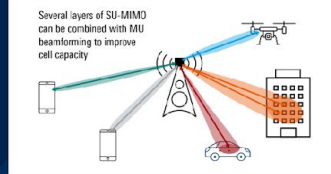
Flexible 5G NR numerology



Flexible 5G network architecture with different virtualization layers



Increasing cell capacity with massive MIMO



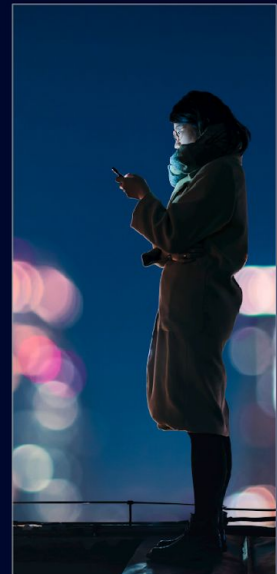
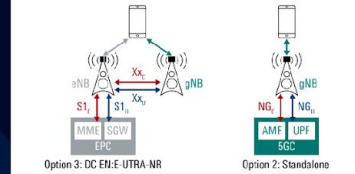
Key parameters in line with 3GPP Rel. 16

Key parameters	FR1	FR2
Carrier aggregation	Up to 16 carriers	
Bandwidth per carrier	5/10/15/20/25/30/40/50/60/70/80/90/100 MHz	50/100/200/400 MHz
Subcarrier spacing	15/30/60 kHz	60/120/240 kHz
Maximum modulation scheme	DL: up to 256QAM UL: up to 4 layers	DL: up to 2 layers UL: up to 2 layers
MIMO scheme	DL: up to 8 layers UL: up to 4 layers	DL: up to 2 layers UL: up to 2 layers
Duplex mode	TDD (f0, f1), FDD	TDD
Access scheme	DL: CP-OFDM, UL: CP-OFDM, DFT-s-OFDM	

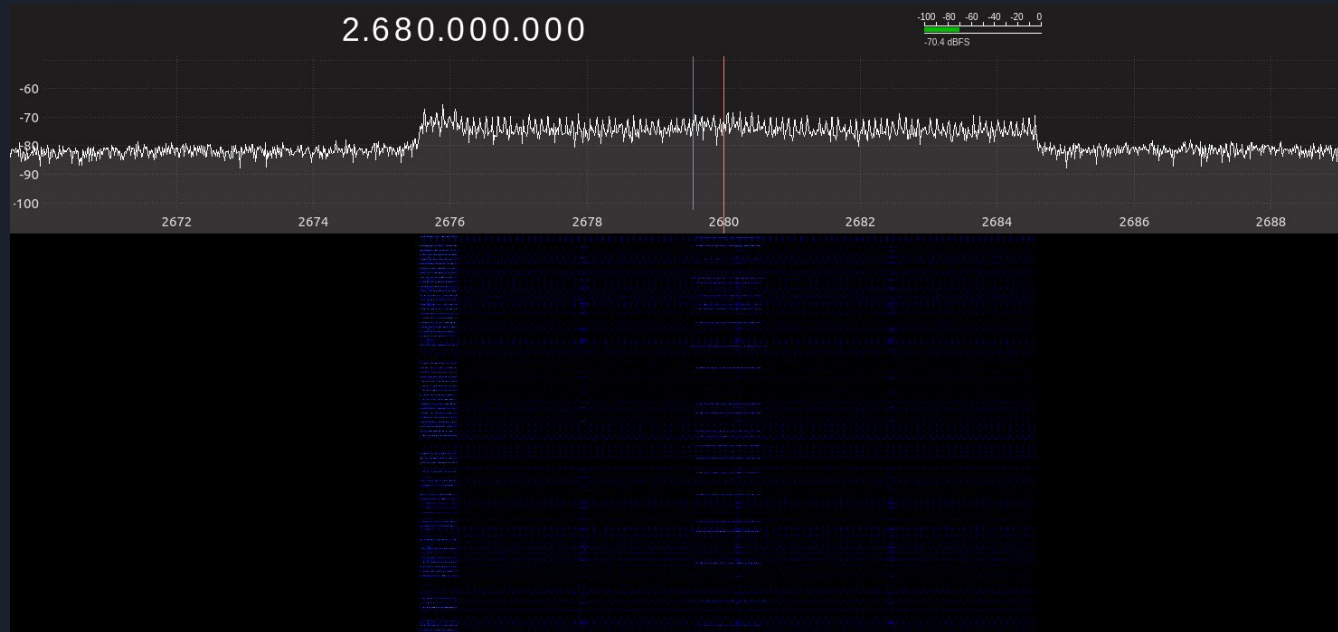
5G NR flexible numerologies

Subcarrier spacing (in kHz)	15	30	60	120	240
Symbol duration (μs)	66.7	33.3	16.7	8.33	4.17
CP duration (μs)	4.7	2.3	1.2 (normal CP) 4.13 (extended CP)	0.59	0.29
Maximum nominal system bandwidth (in MHz)	50	100	200 (FR1) 200 (FR2)	400	400
Maximum FFT size	4096	4096	4096	4096	4096
Symbols per slot	14	14	14 (normal CP) 12 (extended CP)	14	14
Slots per subframe	1	2	4	8	16
Slots per frame	10	20	40	80	160

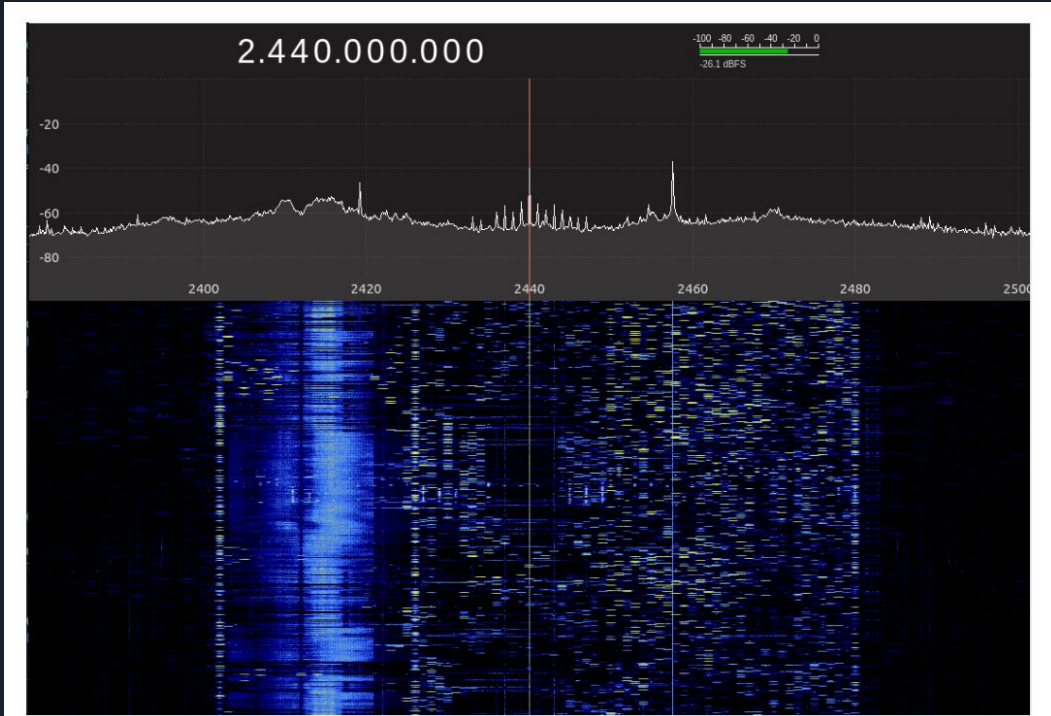
Non-standalone and standalone deployment



Hücresel Teknoloji - 4G



Bluetooth



- 2.4 GHz ISM Band
 - 2.402-2.480 MHz
- Adaptive Freq. Hopping:
 - Classic: 79 x 1MHz
 - LE: 40 x 2 MHz



Daha fazlası

- Atölye çalışması
- Programlama (Python)
- Amatör radyo faaliyetleri
 - Field Day
 - Fox Hunt
- Uydu
 - NOAA
 - ISS
 - QO-100



Decibel Calculation

$$dB = 10 \log_{10} \left(\frac{P_2}{P_1} \right)$$

- A logarithmic unit used to express the ratio of two values
- dB expresses the ratio between two power levels



dB Examples

- +3 dB $\approx$ 2x power
- +10 dB = 10x power
- -3 dB $\approx$ 1/2 power
- -10 dB = 1/10 power



dB Important notes

- What relative to?
 - I am 3 dB taller than my brother
- Amplitude vs. Power
- Gain or Loss
 - 5dB loss
 - -5dB gain
- Add / Subtract
- 3dB around 2x power
- We can calculate all others
 - $\text{Ratio}(0.5) = -3\text{dB}$
 - $\text{Ratio}(1) = 0$
 - $\text{Ratio}(20) = \text{Ratio}(2 \cdot 10) = 3\text{dB} + 10\text{dB} = 13\text{dB}$



Absolute Power Values

$$P_{dBm} = 10 \log_{10} \left(\frac{P_{mW}}{1 \text{ mW}} \right)$$

- dBm is an absolute power level referenced to 1mW
- dBV is relative to 1V
- dBuV is relative to 1uV



dBm Examples

- $0 \text{ dBm} = 1 \text{ mW}$
- $10 \text{ dBm} = 10 \text{ mW}$
- $20 \text{ dBm} = 100 \text{ mW}$
- $-10 \text{ dBm} = 0.1 \text{ mW}$



Incredibly useful in RF

- It allows us to express very large or very small ratios in a manageable way.
- It simplifies calculations: multiplication/division become addition/subtraction.
- Human perception (e.g., hearing, signal strength) is often logarithmic.



An Example Calculation

- Consider a transmit example
 - Output 5dBm
 - Filter has insertion loss of 3dB
 - Amplifier 12dB
 - Cable of 2dB loss
- What is the power at the antenna?
 - $5\text{dBm} - 3\text{dB} + 12\text{dB} - 2\text{dB} = 12\text{dBm}$



Thanks!

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